Result:

batch_size = 64, learning_rate = 0.0001, samples_per_epoch = all, epoch = 100, optim = AdamW

BASE = No augmentation
GAUS = Gaussian noise
PINK = Pink noise
TEMP = Temporal augmentations
SEGR = Segment recombination
CHDO = Channel dropout
MIXU = Mixup-style blending
WGAN = WGAN-GP augmentation
LDIF = LDM augmentation
Composite Augmentation Pipeline (CAP) = PINK + TEMP + CHDO + MIXU
Signal Perturbation Pipeline (CPP) = GAUS + TEMP + SEGR

Model	BASE	GAUS	PINK	TEMP	SEGR	CHDO	MIXU	WGAN	LDIF
EEGNet	0.6781 0.6330	0.6795 0.6295	0.6982 0.6499	0.6858 0.6432	**0.7072** 0.6521	0.6984 0.6459	0.6941 **0.6577**	0.7001 0.6510	0.6993 0.6475
EEGNeX	0.7164 0.6586	0.7191 0.6585	0.7354 0.6757	0.7348 0.6778	**0.7489** **0.6874**	0.7358 0.6601	0.7361 0.6779	0.7443 0.6806	0.7435 0.6821
EEGInceptionERP	0.6568 0.6163	0.6584 0.6245	0.6720 0.6305	0.6683 0.6303	**0.6896** **0.6420**	0.6835 0.6319	0.6780 0.6364	0.6827 0.6359	0.6809 0.6367
CNN-LSTM	0.7145 0.6510	0.7257 0.6595	0.7318 0.6645	0.7300 0.6778	**0.7405** **0.6792**	0.7358 0.6688	0.7352 0.6733	0.7386 0.6761	0.7394 0.6763
EEGConformer	0.6512 0.6214	0.6629 0.6136	0.6784 0.6419	0.6726 0.6455	**0.6862** 0.6490	0.6748 0.6446	0.6779 0.6308	0.6835 **0.6492**	0.6806 0.6489
Average	0.6834 0.6360	0.6891 0.6371	0.7031 0.6525	0.6983 0.6549	**0.7144** **0.6619**	0.7056 0.6502	0.7042 0.6552	0.7098 0.6585	0.7087 0.6583
Δ (%)	0 0	0.83 0.17	2.88 2.59	2.18 2.97	**4.53** **4.07**	3.24 2.23	3.04 3.01	3.86 3.53	3.70 3.50

Table 4.1: Combined comparison of model performance across augmentation settings. Each cell reports Macro F1 (top) and Macro PR AUC (bottom).

Model	BASE	CAP	CPP	CAP + WGAN	CPP + WGAN	CAP + LDIF	CPP + LDIF
EEGNet	0.6781 0.6330	0.7125 0.6609	0.7089 0.6526	0.7160 0.6658	0.7143 0.6658	**0.7175** 0.6614	0.7128 0.6598
EEGNeX	0.7164 0.6586	0.7543 0.6846	0.7498 0.6873	**0.7555** **0.6896**	0.7486 0.6658	0.7550 0.6860	0.7510 0.6831
EEGInceptionERP	0.6568 0.6163	0.6936 0.6487	0.6907 0.6466	**0.6977** 0.6513	0.6954 0.6507	0.6924 0.6543	0.6927 **0.6612**
CNN-LSTM	0.7145 0.6510	0.7491 0.6845	0.7492 0.6824	0.7481 **0.6901**	**0.7503** 0.6880	0.7481 0.6857	0.7463 0.6828
EEGConformer	0.6512 0.6214	0.6920 0.6548	0.6919 0.6536	0.6973 **0.6589**	**0.6954** 0.6572	0.6944 0.6561	0.6920 0.6509
Average	0.6834 0.6360	0.7203 0.6667	0.7181 0.6645	**0.7229** **0.6711**	0.7208 0.6655	0.7214 0.6687	0.7189 0.6675
Δ (%)	0 0	5.39 4.82	5.07 4.48	**5.77** **5.51**	5.47 4.63	5.56 5.14	5.19 4.95

Table 4.2: Combined comparison of model performance. Each cell reports Macro F1 (top) and Macro PR AUC (bottom).

MACRO F1	BASE	GAUS	PINK	TEMP	SEGR	CHDO	MIXU	WGAN	LDIF
EEGNet	0.6781	0.6795	0.6982	0.6858	0.7072	0.6984	0.6941	0.7001	0.6993
EEGNeX	0.7164	0.7191	0.7354	0.7348	0.7489	0.7358	0.7361	0.7443	0.7435
EEGInceptionERP	0.6568	0.6584	0.6720	0.6683	0.6896	0.6835	0.6780	0.6827	0.6809
CNN-LSTM	0.7145	0.7257	0.7318	0.7300	0.7405	0.7358	0.7352	0.7386	0.7394
EEGConformer	0.6512	0.6629	0.6784	0.6726	0.6862	0.6748	0.6779	0.6835	0.6806
Average	0.6834	0.6891	0.7031	0.6983	0.7144	0.7056	0.7042	0.7098	0.7087
Δ (%)	0	0.83	2.88	2.18	4.53	3.24	3.04	3.86	3.70

MACRO PR AUC	BASE	GAUS	PINK	TEMP	SEGR	CHDO	MIXU	WGAN	LDIF
EEGNet	0.6330	0.6295	0.6499	0.6432	0.6521	0.6459	0.6577	0.6510	0.6475
EEGNeX	0.6586	0.6585	0.6757	0.6778	0.6874	0.6601	0.6779	0.6806	0.6821
EEGInceptionERP	0.6163	0.6245	0.6305	0.6303	0.6420	0.6319	0.6364	0.6359	0.6367
CNN-LSTM	0.6510	0.6595	0.6645	0.6778	0.6792	0.6688	0.6733	0.6761	0.6763
EEGConformer	0.6214	0.6136	0.6419	0.6455	0.6490	0.6446	0.6308	0.6492	0.6489
Average	0.6360	0.6371	0.6525	0.6549	0.6619	0.6502	0.6552	0.6585	0.6583
Δ (%)	0	0.17	2.59	2.97	4.07	2.23	3.01	3.53	3.50

MACRO F1	BASE	CAP	CPP	CAP + WGAN	CPP+ WGAN	CAP + LDIF	CPP + LDIF
EEGNet	0.6781	0.7125	0.7089	0.7160	0.7143	0.7175	0.7128
EEGNeX	0.7164	0.7543	0.7498	0.7555	0.7486	0.7550	0.7510
EEGInceptionERP	0.6568	0.6936	0.6907	0.6977	0.6954	0.6924	0.6927
CNN-LSTM	0.7145	0.7491	0.7492	0.7481	0.7503	0.7481	0.7463
EEGConformer	0.6512	0.6920	0.6919	0.6973	0.6954	0.6944	0.6920
Average	0.6834	0.7203	0.7181	0.7229	0.7208	0.7214	0.7189
Δ (%)	0	5.39	5.07	5.77	5.47	5.56	5.19

MACRO PR AUC	BASE	CAP	CPP	CAP + WGAN	CPP+ WGAN	CAP + LDIF	CPP + LDIF
EEGNet	0.6330	0.6609	0.6526	0.6658	0.6658	0.6614	0.6598
EEGNeX	0.6586	0.6846	0.6873	0.6896	0.6658	0.6860	0.6831
EEGInceptionERP	0.6163	0.6487	0.6466	0.6513	0.6507	0.6543	0.6612
CNN-LSTM	0.6510	0.6845	0.6824	0.6901	0.6880	0.6857	0.6828
EEGConformer	0.6214	0.6548	0.6536	0.6589	0.6572	0.6561	0.6509
Average	0.6360	0.6667	0.6645	0.6711	0.6655	0.6687	0.6675
Δ (%)	0	4.82	4.48	5.51	4.63	5.14	4.95

BASE

MACRO F1	seed = 42	seed = 43	seed = 44	Avg.
EEGNet	0.6780	0.6763	0.6801	0.6781
EEGNeX	0.7167	0.7148	0.7178	0.7164
EEGInceptionERP	0.6551	0.6561	0.6592	0.6568
CNN-LSTM	0.7134	0.7127	0.7175	0.7145
EEGConformer	0.6490	0.6536	0.6511	0.6512
MACRO PR AUC	seed = 42	seed = 43	seed = 44	Avg.
EEGNet	0.6346	0.6333	0.6313	0.6330
EEGNeX	0.6554	0.6626	0.6578	0.6586
EEGInceptionERP	0.6159	0.6158	0.6173	0.6163
CNN-LSTM	0.6495	0.6521	0.6514	0.6510
EEGConformer	0.6241	0.6234	0.6168	0.6214

GAUS

MACRO F1	seed = 42	seed = 43	seed = 44	Avg.
EEGNet	0.6747	0.6835	0.6805	0.6795
EEGNeX	0.7184	0.7188	0.7202	0.7191
EEGInceptionERP	0.6545	0.6597	0.6611	0.6584
CNN-LSTM	0.7266	0.7251	0.7256	0.7257
EEGConformer	0.6630	0.6645	0.6614	0.6629
MACRO PR AUC	seed = 42	seed = 43	seed = 44	Avg.
EEGNet	0.6309	0.6275	0.6301	0.6295
EEGNeX	0.6574	0.6595	0.6586	0.6585
EEGInceptionERP	0.6211	0.6259	0.6267	0.6245
CNN-LSTM	0.6573	0.6618	0.6594	0.6595
EEGConformer	0.6112	0.6164	0.6132	0.6136

PINK

MACRO F1	seed = 42	seed = 43	seed = 44	Avg.
EEGNet	0.7014	0.6980	0.6953	0.6982
EEGNeX	0.7335	0.7334	0.7395	0.7354
EEGInceptionERP	0.6711	0.6730	0.6721	0.6720
CNN-LSTM	0.7309	0.7336	0.7310	0.7318
EEGConformer	0.6792	0.6803	0.6757	0.6784
MACRO PR AUC	seed = 42	seed = 43	seed = 44	Avg.
EEGNet	0.6510	0.6499	0.6489	0.6499
EEGNeX	0.6741	0.6795	0.6737	0.6757
EEGInceptionERP	0.6320	0.6298	0.6299	0.6305
CNN-LSTM	0.6668	0.6641	0.6627	0.6645
EEGConformer	0.6415	0.6430	0.6413	0.6419

TEMP

MACRO F1	seed = 42	seed = 43	seed = 44	Avg.
EEGNet	0.6882	0.6851	0.6843	0.6858
EEGNeX	0.7337	0.7325	0.7382	0.7348
EEGInceptionERP	0.6655	0.6709	0.6685	0.6683
CNN-LSTM	0.7273	0.7317	0.7312	0.7300
EEGConformer	0.6698	0.6743	0.6737	0.6726
MACRO PR AUC	seed = 42	seed = 43	seed = 44	Avg.
EEGNet	0.6401	0.6459	0.6436	0.6432
EEGNeX	0.6804	0.6747	0.6784	0.6778
EEGInceptionERP	0.6309	0.6272	0.6329	0.6303
CNN-LSTM	0.6785	0.6772	0.6779	0.6778
EEGConformer	0.6471	0.6438	0.6458	0.6455

SEGR

MACRO F1	seed = 42	seed = 43	seed = 44	Avg.
EEGNet	0.7051	0.7056	0.7109	0.7072
EEGNeX	0.7490	0.7453	0.7524	0.7489
EEGInceptionERP	0.6922	0.6878	0.6890	0.6896
CNN-LSTM	0.7445	0.7408	0.7362	0.7405
EEGConformer	0.6887	0.6822	0.6877	0.6862
MACRO PR AUC	seed = 42	seed = 43	seed = 44	Avg.
EEGNet	0.6521	0.6488	0.6554	0.6521
EEGNeX	0.6849	0.6905	0.6870	0.6874
EEGInceptionERP	0.6424	0.6407	0.6429	0.6420
CNN-LSTM	0.6788	0.6808	0.6781	0.6792
EEGConformer	0.6508	0.6469	0.6495	0.6490

CHDO

MACRO F1	seed = 42	seed = 43	seed = 44	Avg.
EEGNet	0.6998	0.6952	0.7003	0.6984
EEGNeX	0.7385	0.7331	0.7359	0.7358
EEGInceptionERP	0.6827	0.6848	0.6830	0.6835
CNN-LSTM	0.7351	0.7366	0.7357	0.7358
EEGConformer	0.6747	0.6753	0.6746	0.6748
MACRO PR AUC	seed = 42	seed = 43	seed = 44	Avg.
EEGNet	0.6475	0.6439	0.6464	0.6459
EEGNeX	0.6630	0.6594	0.6579	0.6601
EEGInceptionERP	0.6341	0.6312	0.6306	0.6319
CNN-LSTM	0.6669	0.6720	0.6676	0.6688
EEGConformer	0.6445	0.6464	0.6429	0.6446

MIXU

MACRO F1	seed = 42	seed = 43	seed = 44	Avg.
EEGNet	0.6911	0.6963	0.6949	0.6941
EEGNeX	0.7382	0.7345	0.7356	0.7361
EEGInceptionERP	0.6785	0.6797	0.6759	0.6780
CNN-LSTM	0.7346	0.7384	0.7328	0.7352
EEGConformer	0.6789	0.6765	0.6783	0.6779
MACRO PR AUC	seed = 42	seed = 43	seed = 44	Avg.
EEGNet	0.6548	0.6584	0.6600	0.6577
EEGNeX	0.6784	0.6802	0.6753	0.6779
EEGInceptionERP	0.6374	0.6351	0.6367	0.6364
CNN-LSTM	0.6734	0.6742	0.6723	0.6733
EEGConformer	0.6293	0.6306	0.6325	0.6308

WGAN

MACRO F1	seed = 42	seed = 43	seed = 44	Avg.
EEGNet	0.6988	0.7004	0.7011	0.7001
EEGNeX	0.7446	0.7455	0.7429	0.7443
EEGInceptionERP	0.6805	0.6840	0.6836	0.6827
CNN-LSTM	0.7417	0.7347	0.7395	0.7386
EEGConformer	0.6823	0.6823	0.6861	0.6835
MACRO PR AUC	seed = 42	seed = 43	seed = 44	Avg.
EEGNet	0.6547	0.6497	0.6488	0.6510
EEGNeX	0.6791	0.6834	0.6794	0.6806
EEGInceptionERP	0.6374	0.6342	0.6363	0.6359
CNN-LSTM	0.6764	0.6771	0.6750	0.6761
EEGConformer	0.6486	0.6522	0.6469	0.6492

LDIF

MACRO F1	seed = 42	seed = 43	seed = 44	Avg.
EEGNet	0.6977	0.7020	0.6983	0.6993
EEGNeX	0.7435	0.7446	0.7426	0.7435
EEGInceptionERP	0.6834	0.6803	0.6790	0.6809
CNN-LSTM	0.7411	0.7405	0.7367	0.7394
EEGConformer	0.6809	0.6804	0.6807	0.6806

MACRO PR AUC	seed = 42	seed = 43	seed = 44	Avg.
EEGNet	0.6468	0.6502	0.6455	0.6475
EEGNeX	0.6811	0.6850	0.6804	0.6821
EEGInceptionERP	0.6398	0.6360	0.6343	0.6367
CNN-LSTM	0.6771	0.6775	0.6745	0.6763
EEGConformer	0.6492	0.6459	0.6516	0.6489

BASE

Model	musc F1	eyem F1	elec F1	chew F1	Macro F1
EEGNet	0.7325	0.8108	0.6714	0.4978	0.6781
EEGNeX	0.7509	0.8282	0.7513	0.5353	0.7164
EEGInceptionERP	0.6928	0.7947	0.6870	0.4526	0.6568
CNN-LSTM	0.7472	0.8318	0.7549	0.5242	0.7145
EEGConformer	0.6893	0.7836	0.7019	0.4302	0.6512
Avg	0.7225	0.8098	0.7133	0.4880	0.6834

Model	musc PR AUC	eyem PR AUC	elec PR AUC	chew PR AUC	Macro PR AUC
EEGNet	0.6627	0.7762	0.6012	0.4922	0.6330
EEGNeX	0.6661	0.8037	0.6604	0.5044	0.6586
EEGInceptionERP	0.6233	0.7386	0.5922	0.5113	0.6163
CNN-LSTM	0.6312	0.7844	0.6641	0.5245	0.6510
EEGConformer	0.6156	0.7469	0.6375	0.4856	0.6214
Avg	0.6397	0.7699	0.6310	0.5036	0.6360

Improved

Model	musc F1	eyem F1	elec F1	chew F1	Macro F1
EEGNet	0.7533	0.8312	0.6925	0.5202	0.6993
EEGNeX	0.7713	0.8517	0.7761	0.5749	0.7435
EEGInceptionERP	0.7134	0.8146	0.7156	0.4800	0.6809
CNN-LSTM	0.7689	0.8547	0.7757	0.5586	0.7394
EEGConformer	0.7121	0.8019	0.7285	0.4799	0.6806
Avg	0.7438	0.8308	0.7377	0.5227	0.7087

Model	musc PR AUC	eyem PR AUC	elec PR AUC	chew PR AUC	Macro PR AUC
EEGNet	0.6828	0.7942	0.6285	0.4845	0.6475
EEGNeX	0.6919	0.8231	0.6952	0.5182	0.6821
EEGInceptionERP	0.6457	0.7593	0.6151	0.5267	0.6367
CNN-LSTM	0.6591	0.8059	0.6958	0.5444	0.6763
EEGConformer	0.6418	0.7682	0.6765	0.5091	0.6489
Avg	0.6643	0.7901	0.6622	0.5166	0.6583